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AI in Construction and Facilities Management
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SUMMARY:

1. Main Findings and Arguments

The article presents a framework that integrates **Building Information Modeling (BIM)**, **Large Language Models (LLMs)**, and the **Model Context Protocol (MCP)** to automate construction cost estimation. The system connects BIM models from Autodesk Revit with industry cost databases, allowing AI to automatically extract quantities, match components with cost items, and generate professional estimates.

The case study demonstrated significant efficiency improvements. While traditional manual estimation took approximately **2.5–3.5 hours**, the automated system completed the same task in about **42 seconds**, representing an efficiency improvement of roughly **98%**. The automated estimate also achieved results close to professional standards, with only a **5.1% difference compared to manual estimates**, showing that AI-driven workflows can provide both speed and reasonable accuracy in construction cost estimation.

2. Broader Implications for Society

The research highlights the growing role of **artificial intelligence in improving productivity and digital transformation in the construction industry**, which historically has lagged behind other industries in adopting new technologies. By automating complex tasks such as quantity take-offs and cost calculations, AI systems can reduce human errors, improve decision-making, and accelerate project planning processes.

These advancements can help construction firms deliver projects more efficiently and potentially reduce project costs and delays. Additionally, the use of conversational AI interfaces allows professionals without programming expertise to interact with complex systems using natural language, making advanced technologies more accessible to industry practitioners.

3. Relevance and Potential Influence on the Construction Industry

The study demonstrates how integrating **AI, BIM, and standardized communication protocols like MCP** can transform traditional construction workflows. By automating the full cost estimation pipeline—from model interrogation to report generation—the framework shows the potential for a more **connected, intelligent, and scalable construction process**.

If widely adopted, this approach could significantly improve productivity, reduce estimation time, and standardize cost estimation practices across the industry. It also provides a blueprint for future research and development in construction automation, encouraging further integration of AI tools with existing engineering and project management software.